MILITARY SPECIFICATION

CABLE ASSEMBLY, SPECIAL PURPOSE, ELECTRICAL: T-4087

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of cable, covering a series of six assemblies entitled Cable Assembly, Special Purpose, Electrical: T-4087-2; T-4087-5; T-4087-6; T-4087-7; T-4087-8; and T-4087-12 which are used in the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8800545	- Bag, Packing, Ammunition, For Cable Assembly.
8800546	- Bag, Packing, Ammunition, For Cable Assembly.
8800716	- Connector.
8800850	- Cable Assembly; T-4087-5.
8808362	- Container, Packing, Ammunition, For Domestic (CONUS) Shipments of Special Weapons Items, Packing and Marking.
8813402	- Cable Assembly; T-4087-12.
8813403	- Cable Assembly; T-4087-2.
8813597	- Cable Assembly; T-4087-8.
8813599	- Cable Assembly; T-4087-6.
8813600	- Cable Assembly; T-4087-7.
8833932	- Case, Combination.
EL-8842546	- Equipment List.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The cables shall comply with all requirements specified on the drawings (dwgs) 8800850, 8813402, 8813403, 8813597, 8813599, and 8813600, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Mating assembly. The cable plugs shall be capable of being fully mated and unmated to a receptacle by hand, without use of tools, when tested as specified in 4.4.1.

3.5 Continuity. The direct current resistance of each circuit of the cable assembly shall not exceed the value specified for the respective circuit in Tables I through VI, when measured as specified in 4.4.2.

Table I

T-4087-2 Continuity

Refer to dwgs 8800716 and 8813403 for connector identification.

From	To	Max Resistance, Ohms	From	To	Max Resistance, Ohms
P1-1	P2-1	0.15	P2-37	P2-20	0.05
P1-2	P2-2	0.15	P1-38	P2-38	0.15
P1-3	P2-3	0.15	P1-39	P2-39	0.15
P1-4	P2-4	0.15	P1-40	P2-40	0.15
P1-6	P2-6	0.15	P1-41	P2-41	0.15
P1-7	P2-7	0.15	P1-42	P1-43	0.05
P1-8	P2-8	0.15	P1-42	P1-44	0.05
P1-9	P2-9	0.15	P2-42	P2-43	0.05
P1-10	P2-10	0.15	P2-42	P2-44	0.05
P1-11	P2-11	0.15	P1-45	P2-45	0.15
P1-12	P2-12	0.15	P1-46	P2-46	0.15
P1-13	P2-13	0.15	P1-47	P2-48	0.05
P1-13	P1-14	0.05	P1-47	P1-49	0.05
P1-13	P1-5	0.05	P2-47	P2-48	0.05
P1-13	P2-14	0.15	P2-47	P2-49	0.05

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From	To	Max Resistance, Ohms	From	To	Max Resistance, Ohms
P1-13	P2-5	0.15	P1-50	P2-50	0.15
P1-15	P2-15	0.15	P1-51	P2-51	0.15
P1-16	P2-16	0.15	P1-52	P2-52	0.15
P1-17	P2-17	0.15	P1-53	P2-54	0.05
P1-18	P2-18	0.15	P2-53	P2-55	0.05
P1-19	P2-19	0.15	P1-55	P2-55	0.15
P1-21	P2-21	0.15			
P1-22	P2-22	0.15			
P1-23	P2-23	0.15			
P1-24	P2-24	0.15			
P1-25	P2-25	0.15			
P1-26	P2-26	0.15			
P1-27	P2-27	0.15			
P1-28	P2-28	0.15			
P1-29	P2-29	0.15			
P1-30	P2-30	0.15			
P1-31	P2-31	0.15			
P1-32	P2-32	0.15			
P1-33	P2-33	0.15			
P1-34	P2-34	0.15			
P1-35	P2-35	0.15			
P1-36	P2-36	0.15			
P1-37	P1-20	0.05			

Table II

T-4087-5 Continuity

Refer to dwg 8800850 for connector identification

From	To	Maximum Resistance Ohms
P1-A	P1-B	0.05
P1-A	P2-A	0.05
P1-A	P2-B	0.05
P1-C	P2-C	0.05

Table III

T-4087-6 Continuity

Refer to dwg 8813599 for connector identification

From	To	Maximum Resistance Ohms
P1-A	P2-A	0.05
P1-B	P1-C	0.05
P1-B	P1-D	0.05
P1-B	P2-D	0.05
P1-B	P2-C	0.05

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From	To	Maximum Resistance Ohms
P1-B	P2-B	0.05
P1-E	P1-F	0.05
P1-E	P1-G	0.05
P1-E	P2-G	0.05
P1-E	P2-F	0.05
P1-E	P2-E	0.05

Table IV

T-4087-7 Continuity

Refer to dwg 8813600 for connector identification

From	To	Maximum Resistance Ohms
P1-A	P2-A	0.05
P1-B	P1-C	0.05
P1-B	P1-D	0.05
P1-B	P2-D	0.05
P1-B	P2-C	0.05
P1-B	P2-B	0.05
P1-E	P1-F	0.05
P1-F	P1-G	0.05
P1-F	P2-G	0.05
P1-F	P2-F	0.05
P1-F	P2-E	0.05

Table V

T-4087-8 Continuity

Refer to dwg 8813597 for connector identification

From	To	Maximum Resistance Ohms
P2-A	P1-A	0.05
P2-B	P2-E	0.05
P2-C	P2-D	0.05
P2-D	P2-L	0.05
P2-F	P2-G	0.05
P2-H	P1-L	0.05
P2-H	P1-M	0.05
P2-J	P1-I	0.05
P2-J	P1-N	0.05
P2-K	P1-H	0.05
P2-M	P1-G	0.05
P2-M	P1-F	0.05
P1-B	P1-C	0.05
P1-B	P1-E	0.05
P1-D	P1-J	0.05

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Table VI

T-4087-12 Continuity

Refer to dwg 8813402 for connector identification

From	To	Maximum Resistance Ohms
P1-A	P2-A	0.10
P1-B	P2-B	0.10
P1-C	P2-C	0.10

3.6 Insulation resistance. The insulation resistance of each terminal to all other terminals listed for the respective assemblies in Tables VII through XII shall be greater than 20 megohms when tested at 500 ± 50 volts direct current (vdc) as specified in 4.4.3.

Table VII

T-4087-12 Insulation resistance

Refer to dwg 8813402 for connector identification

Terminal	
P1-A	
P1-B	
P1-C	
GROUND	- This will be the test fixture ground, P2 cable connector shells shall be at ground potential.

Table VIII

T-4087-6 Insulation resistance

Terminal	
P1-A	
P1-B	
P1-E	
P1-Unused terminals H,J,K	- Note 1
P2-Unused terminals H,J,K	- Note 1
Ground	- This will be the test fixture ground - Note 2

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Note 1: Unused terminals shall be connected together for this test and tested as one terminal.
Note 2: Cable connector shells shall be at ground potential.
Refer to dwg 8813599 for connector identification.

Table IX

T-4087-7 Insulation resistance

Terminals
P1-A
P1-B
P1-E
P1-Unused terminals H,J,K - Note 1
P2-Unused terminals H,J,K - Note 1
Ground - This will be the test fixture ground - Note 2

Note 1: Unused terminals shall be connected together for this test and tested as one terminal.
Note 2: Cable connector shells shall be at ground potential.
Refer to dwg 8813600 for connector identification.

Table X

T-4087-8 Insulation resistance

Terminals
P1-A
P1-B
P1-D
P1-F
P1-H
P1-I
P1-K
P1-L
P2-B
P2-C
P2-F
P2-Unused terminals I,N - Note 1
Ground - This will be the test fixture ground - Note 2

Note 1: Unused terminals shall be connected together for this test and tested as one terminal.
Note 2: Cable connector shells shall be at ground potential.
Refer to dwg 8813597 for connector identification.

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Table XI

T-4087-5 Insulation resistance

Terminal
P1-A
P1-C
Ground - This will be the test fixture ground - Note.

Refer to dwg 8800850 for connector identification.
Note: Cable connector shells shall be at ground potential.

Table XII

T-4087-2 Insulation resistance

Terminals		
P1-1	P1-21	P1-40
P1-2	P1-22	P1-41
P1-3	P1-23	P1-42
P1-4	P1-24	P1-45
P1-5	P1-25	P1-46
P1-6	P1-26	P1-47
P1-7	P1-27	P1-50
P1-8	P1-28	P1-51
P1-9	P1-29	P1-52
P1-10	P1-30	P1-53
P1-11	P1-31	P1-55
P1-12	P1-32	P2-42
P1-15	P1-33	P2-20
P1-16	P1-34	P2-47
P1-17	P1-35	P2-53
P1-18	P1-36	Ground - This will be the test fixture ground.
P1-19	P1-38	Cable connector shells shall be at ground potential.
P1-20	P1-39	

Refer to dwg 8813403 for connector identification.

3.7 Environmental testing. The cables shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The cables shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.4. Tests shall be conducted in the order specified.

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3.7.1 High temperature. The cable assembly shall withstand exposure to a temperature of 160 degrees Fahrenheit (°F) plus or minus (±) 5°F without damage or deterioration and the cable assembly shall comply with 3.4, 3.5 and 3.6 while at 160 ± 5°F when tested as specified in 4.4.4.1.

3.7.2 Low temperature. The cable assembly shall withstand exposure to a temperature of -80 ± 5°F without evidence of damage or deterioration and shall comply with 3.4, 3.5 and 3.6 while at a temperature of -65 ± 5°F when tested as specified in 4.4.4.2.

3.7.3 Temperature shock. The cable assemblies shall withstand temperature shock of 160 ± 5°F to 80 ± 20°F to -65 ± 5°F without damage or deterioration when tested as specified in 4.4.4.3. Upon completion of this test the cable assemblies shall comply with 3.5 and 3.6.

3.7.4 Cold bend.

3.7.4.1 Cable assemblies T-4087-5, T-4087-6, T-4087-7 and T-4087-8. The insulation shall not crack, separate from the connector, or otherwise deteriorate when subjected to a temperature of -40 ± 5°F and flexed as specified in 4.4.4.1. Upon completion of this test the cable assemblies shall comply with 3.5 and 3.6.

3.7.4.2 Cable assemblies T-4087-2 and T-4087-12. The cable assemblies shall show no evidence of visible cracks or other deterioration when subjected to a temperature of -40 ± 5°F and flexed as specified in 4.4.4.4.2. Upon completion of this test the cable assemblies shall comply with 3.5 and 3.6.

3.7.5 Humidity. The cable assembly shall withstand conditions of 95 ± 5 percent relative humidity and condensation at temperatures cycling from 80 to ± 160°F without damage when tested as specified in 4.4.4.5.

3.8 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460(MU).

3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.9.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 5 complete assemblies (see dwgs 8800850, 8813402, 8813403, 8813597, 8813599 and 8813600) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table XIII. Tests shall be conducted in order specified in Table XIII.

Table XIII

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Mating assembly (see 3.4)		X
3. Continuity (see 3.5)		X
4. Insulation resistance (see 3.6)		X
5. High temperature (see 3.7.1)		X
6. Low temperature (see 3.7.2)		X
7. Temperature shock (see 3.7.3)	X	
8. Cold bend (see 3.7.4)		X
9. Humidity (see 3.7.5)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

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4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table XIV and Table XV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table XIV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table XV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cable assemblies, T-4087-2, T-4087-5, T-4087-6, T-4087-7, T-4087-8, and T-4087-12 prior to potting (see dwgs 8800850, 8813402, 8813403, 8813597, 8813599 and 8813600 as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper connector	Visual
	102. Improper soldering (see 3.8)	Visual
	103. Evidence of poor workmanship (see 3.9)	Visual
	104. Any component missing, damaged or loose	Visual
Minor:	None defined.	

4.3.2.1.2 Cable assemblies, T-4087-2, T-4087-5, T-4087-6, T-4087-7, T-4087-8, and T-4087-12 (see dwgs 8800850, 8813402, 8813403, 8813597, 8813599, and 8813600 as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Length improper	Gage
	102. Improper location of male terminals (T-4087-2 cable only)	Gage
	103. Any pin bent	Visual

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104.	Any pin missing (as evidenced by empty hole in insert)	Visual
105.	Deformation of connector sufficient to impair mating	Visual
106.	Any component cracked	Visual
107.	Portion of connector broken away (minor chips or gouges which do not affect mating are not objectionable)	Visual
108.	Potting on face of inserts	Visual
109.	Voids or air pockets in potting greater than 1/8 inch in diameter	Visual
110.	Leadwire(s) not completely covered by potting	Visual
111.	Permanently damaged terminal(s) (gouged, nicked or flattened)	Visual
*112.	Height of pins (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins)	Visual
**113.	Depth of sockets (female sockets of the same diameter must all appear to be within 1/16 inch in depth relative to the adjacent sockets)	Visual
114.	Cracks in potting	Visual
115.	Any component missing or insecurely assembled	Visual/Manual
116.	Objective evidence relative to material missing or improper (see 3.1)	Visual

Minor:

201.	Marking, mislocated, missing or incorrect	Visual
202.	Poor adhesion of potting to connectors	Visual
203.	Evidence of poor workmanship (see 3.9)	Visual

*In the event pins do not appear to be within 1/16 inch in height relative to the adjacent pins, a gage (see 4.3.4.1) shall be used to determine that variation from the normal height. Reject any pin that varies by more than 0.063 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward edge of the connector shall be utilized as the reference surface for all measurements of pin heights.

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**In the event socket inserts do not appear to be within 1/16 inch in depth relative to adjacent sockets, a gage (see 4.3.4.1) shall be used to determine that variation from the normal depth. Reject any insert that varies by more than 0.063 inch maximum from the normal socket depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward edge of the connector shall be utilized as a reference surface for all measurements of socket depths.

4.3.2.1.3 Bag, packing, ammunition, for cable assembly (see dwgs 8800545 and 8800546 as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Connector envelope missing or incorrect	Visual
102.	Wrong size bag	Visual
103.	Bag not sealed	Visual
104.	Marking missing, mislocated or incorrect	Visual
105.	Air in bag not minimized before sealing	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.9)	Visual

4.3.2.1.4 Packing, container (see dwg 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Container damaged to extent that contents show	Visual
102.	Marking missing or incorrect	Visual
103.	Contents loose	Manual
Minor:		
201.	Evidence of poor workmanship (see 3.9)	Visual

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Continuity	Major
b. Insulation resistance	Major
c. Cold bend	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Continuity and insulation resistance. Each cable shall be tested in accordance with 4.4.2 and 4.4.3. If any item fails to comply with 3.5 and 3.6, the item shall be rejected and removed from the lot.

4.3.3.2 Cold bend. A sample shall be selected at random in accordance with Table XVI. The sample shall be tested in accordance with 4.4.4.4. If any item fails to comply with 3.7.4, the lot shall be rejected.

Table XVI

Lot Size	Sample Size
3-50	3
51-150	4
151-300	5
301-600	6
601-1000	8
1001-and over	MIL-STD-105(AQL 0.25%)

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8842546). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

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4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Mating assembly. The cable assemblies shall be mounted by hand to a fixture having mating receptacles for the cable plugs. Care must be exercised to avoid bending connector terminals.

4.4.2 Continuity. The resistance between the terminals specified shall be determined by the use of a suitable electrical resistance measuring instrument. Requirements shall be as specified in 3.5.

4.4.3 Insulation resistance. The insulation resistance shall be determined by the use of an insulation resistance tester capable of supplying the required voltage and having a scale reading directly in megohms. The current shall be limited to a maximum of 5 milliamperes to protect human life. Requirements shall be as specified in 3.6.

4.4.4 Environmental testing (see 3.7).

4.4.4.1 High temperature. The cable assembly shall be placed within the chamber and the internal temperature of the chamber shall be raised to 160 ± 5°F. The cable assembly shall be maintained at 160 ± 5°F for a period of 2 hours. While still at this temperature, the cable assembly shall be tested as specified in 4.4.1 through 4.4.3.

4.4.4.2 Low temperature. The cable assembly shall be placed within the chamber and the chamber cooled to, and maintained at a temperature of -80 ± 5°F for a period of 4 hours. The temperature of the chamber shall then be raised to -65 ± 5°F and maintained for an additional 2 hour period. At the conclusion of this exposure period and while at this temperature, the cable assembly shall be tested as specified in 4.4.1 through 4.4.3.

4.4.4.3 Temperature shock. The cable assembly shall be placed within a test chamber where the temperature is 160 ± 5°F. The cable assembly shall be subjected to this temperature for a period of 4 hours, at the conclusion of which, and within 5 minutes, the cable assembly shall be transferred to a chamber having an internal temperature of 80 ± 20°F. The cable assembly shall be subjected to this temperature for a

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period of 4 hours minimum (min), after which the cable assembly shall be transferred to a chamber having an internal temperature of -65 ± 5°F. The cable assembly shall be subjected to this temperature for a period of 4 hours min. At the conclusion of this test and within 5 minutes, the cable assembly shall be transferred to the 80 ± 20°F test chamber and the cable assembly shall be subjected to this temperature for a min of 4 hours. This completes one cycle, the number of complete cycles shall be three. At the completion of the third cycle the cable assembly shall be removed from the test chamber and tested as specified in 4.4.2 and 4.4.3.

4.4.4.4 Cold bend.

4.4.4.4.1 Cable assemblies T-4087-5, T-4087-6, T-4087-7, and T-4087-8. The cable assembly shall be placed within a test chamber and the temperature of the chamber lowered to -40 ± 5°F. The assembly shall remain at this temperature for a minimum of 4 hours. At the conclusion of this period and while at this temperature the cable assembly shall be tested. (see Figure 1). With connector A free to rotate around its own axis, rotate connector B about its own axis three times at 20 ± 4 revolutions per minute. Distance, d, for each connector assembly shall be as shown on Figure 1. After this test the assembly shall be tested as specified in 4.4.2 and 4.4.3.

4.4.4.4.2 Cable assemblies T-4087-2 and T-4087-12. The cold bend test on the cable assemblies shall consist of coiling the assemblies in the storage compartment of the combination case (dwg 8833932), and placing the case within a test chamber for a minimum of 4 hours at -40 ± 5°F. While at this temperature the cable assemblies shall be removed from the case, straightened, and replaced in the case. The case and cables may be removed from the test chamber for 3 minute periods during handling provided they are returned and their temperature stabilized (see 6.4) between handling periods. After the cables have returned to normal room temperature and have completely dried they must show no signs of cracking and shall be tested as specified in 4.4.2 and 4.4.3.

4.4.4.5 Humidity. The cable assembly shall be placed in the test chamber and set up to simulate installed conditions. The chamber temperature shall be 80 ± 20°F. During the first 2 hour period, the temperature shall be gradually raised to 160 ± 5°F. This temperature shall be maintained during the next 6 hour period. During the following 16 hour period, the temperature in the chamber shall be gradually reduced to 80 ± 20°F which constitutes one cycle. The relative humidity throughout the cycle shall be 95 ± 5 percent. The cycle shall be repeated 10 times. Distilled or demineralized water having a pH value of between 6.5 and 7.5 shall be used. The velocity through the test area shall not exceed 150 feet per minute. Upon completion of the tenth cycle the assembly shall be removed immediately from the test chamber

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and excessive moisture removed by blowing dry, filtered, compressed air over the unit for a period of 5 minutes. The assembly shall then be tested as specified in 4.4.2 and 4.4.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The cables shall be packaged in accordance with dwgs 8800545 or 8800546 as applicable using the specified moisture barrier.

5.1.2 Level C. The cables shall be packaged in accordance with dwgs 8800545 or 8800546 as applicable except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The cables shall be packed in accordance with dwg 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwgs 8800545 or 8800546, and 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

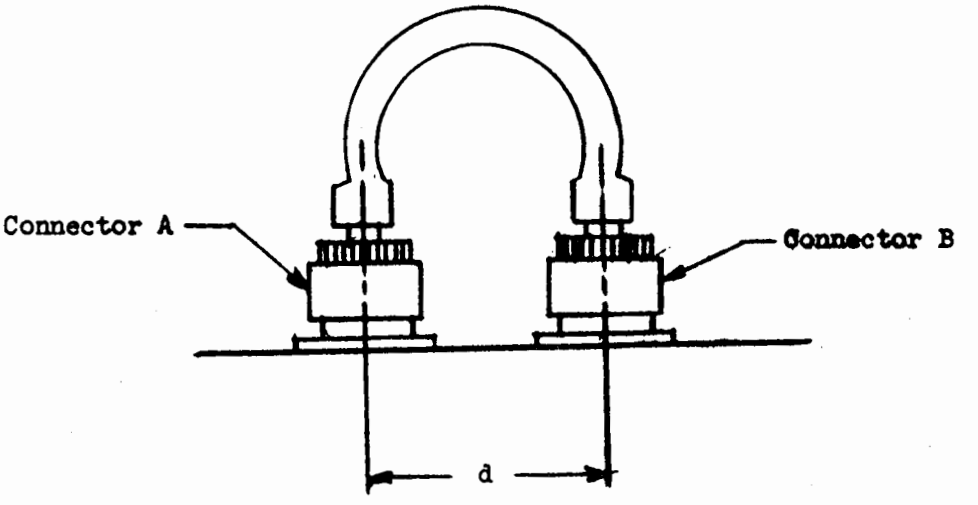
6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1687 dated 19 September 1961.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A070

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Cable Ass'y	d (in inches)
T4087-5	1 1/2 ± 1/8
T4087-6	3.0 ± 1/4
T4087-7	1 3/8 ± 1/8
T4087-8	13.0 ± 1/4

Figure 1

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